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SECP3213-01 - BUSINESS INTELLIGENCE

Phase 2: Implementation (Alteryx Workflow)

Group: Klever

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Introduction

This phase focuses on utilizing Alteryx Designer to construct the Business Intelligence workflow that was suggested in Phase 1. Preparing, cleaning, transforming, and analyzing datasets from three major streaming services Netflix, Amazon Prime Video, and Disney+ is the goal of this phase.

The project objectives, which include comparing content availability across platforms, determining geographic content distribution, assessing genre patterns, looking at content trends over time, and researching audience rating distributions, were supported by the workflow. In Phase 3, Power BI will be used to view the processed datasets.

Workflow Overview

The Alteryx workflow consists of four major stages:

1. Data Input
2. Data Cleaning
3. Data Transformation
4. Data Analysis

The workflow begins by importing the datasets from Netflix, Amazon Prime Video, and Disney+. Data cleaning and transformation processes are then performed before the data is divided into six analytical branches for further analysis.

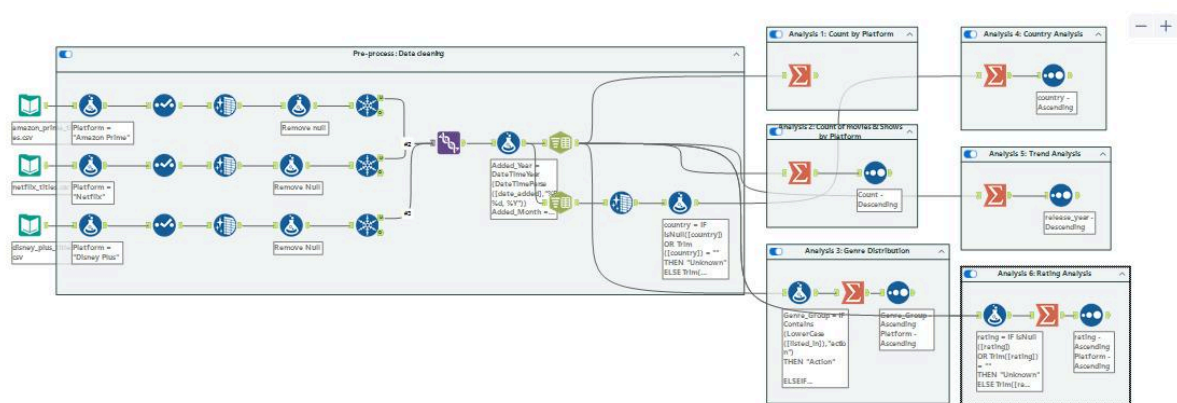


Figure 1: Overall Alteryx Workflow

The figure above illustrates the complete Alteryx workflow used in this project. The workflow integrates data preparation and analysis processes into a single workflow that generates multiple outputs for business intelligence reporting.

Data Input and Integration

Three Input Data tools were used to import the datasets into Alteryx Designer.

The datasets used are:

- Netflix Titles Dataset
- Amazon Prime Video Dataset
- Disney+ Titles Dataset

A Formula Tool was applied to each dataset to create a new attribute called Platform. This attribute identifies the source platform of each record.

Title	Platform
Wednesday	Netflix
Reacher	Amazon Prime
Frozen	Disney+

Field names were standardized and consistency across all datasets was ensured using the Select Tool. The Union Tool was used to merge the standardized datasets into a single, cohesive dataset.

While maintaining platform identification, the integration method enables material from all three platforms to be examined together.

Data Cleaning

Data cleaning was performed to improve data quality and consistency before analysis.

Several cleaning activities were conducted:

- Removal of unnecessary spaces.
- Standardization of data values.

- Handling missing values.
- Replacement of null values with meaningful labels.

Missing values in important fields such as director, cast, country, and rating were replaced with the value “Unknown” to avoid incomplete analytical results.

Data Transformation

Several transformations were performed to prepare the data for analysis.

A Formula Tool was used to create additional fields that support analytical processing.

- Added_Year

The Added_Year field was created to extract the year from the date_added field.

Example:

September 9, 2020 → 2020

- Added_Month

The Added_Month field was created to extract the month from the date_added field.

Example:

September 9, 2020 → 9

These attributes support trend analysis and time-based visualizations.

Analysis 1: Count by Platform

The first analysis was conducted to determine the total number of titles available on each streaming platform.

Method:

- Group By Platform
- Count Title

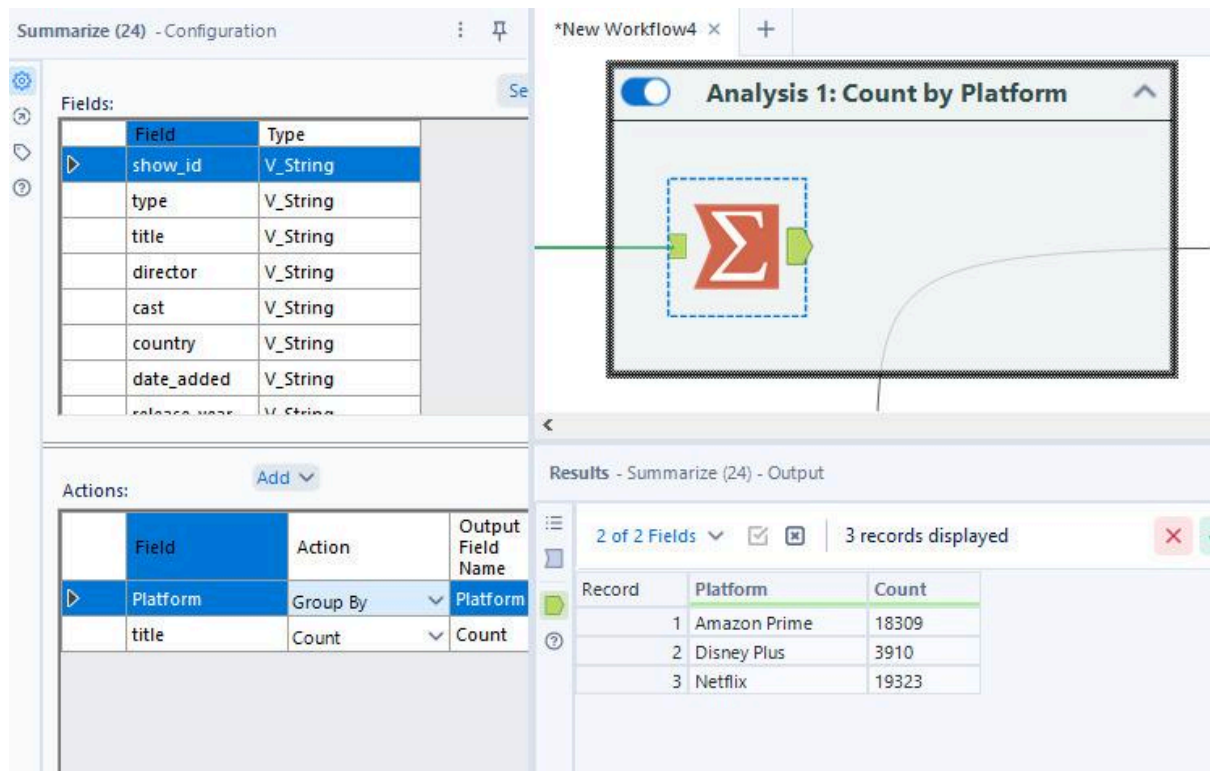


Figure 2 : Analysis 1 – Count by Platform

Result:

Platform	Count
Netflix	19,323
Amazon Prime	18.309
Disney+	3,910

Interpretation:

Netflix contains the largest content library among the three platforms, followed closely by Amazon Prime Video. Disney+ has a smaller but more focused content collection.

Analysis 2: Count of Movies and TV Shows by Platform

This analysis compares the number of movies and television shows available on each streaming platform.

Method:

- Group By Platform
- Group By Type
- Count Title

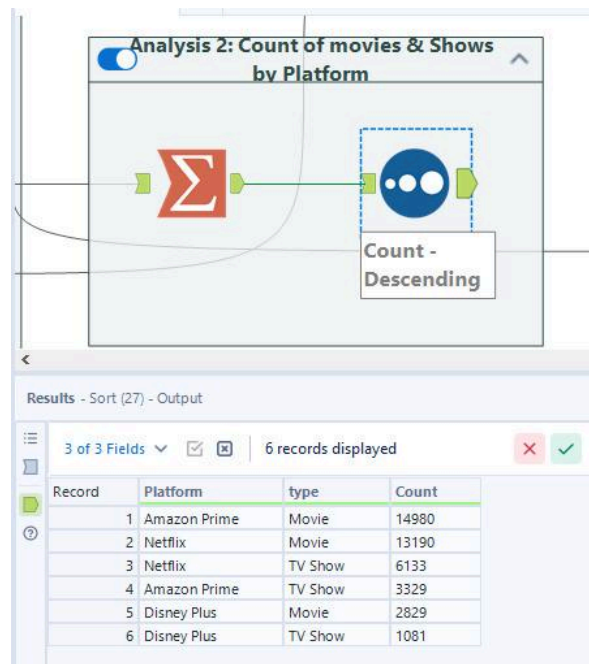


Figure 3 : Analysis 2 – Count of Movies and TV Shows by Platform

Result:

Platform	Type	Count
Amazon Prime	Movie	14,980
Amazon Prime	TV shows	3,329
Netflix	Movie	13,190
Netflix	TV Show	6,133

Disney+	Movie	2,829
Disney+	TV Show	1,081

Interpretation:

Movies dominate all three platforms. Netflix contains a significantly higher number of television shows compared to Amazon Prime Video and Disney+, indicating a stronger emphasis on episodic content.

Analysis 3: Genre Distribution

This analysis examines the distribution of genres across the three streaming platforms.

The genre information was processed and categorized to improve consistency and comparison across platforms.

Method:

- Genre categorization
- Group By Platform
- Group By Genre
- Count Title

Analysis 3: Genre Distribution

Results - Sort (27) - Output

3 of 3 Fields 6 records displayed

Record	Platform	type	Count
1	Amazon Prime	Movie	14980
2	Netflix	Movie	13190
3	Netflix	TV Show	6133
4	Amazon Prime	TV Show	3329
5	Disney Plus	Movie	2829
6	Disney Plus	TV Show	1081

Figure 4: Analysis 3 – Genre Distribution

Interpretation:

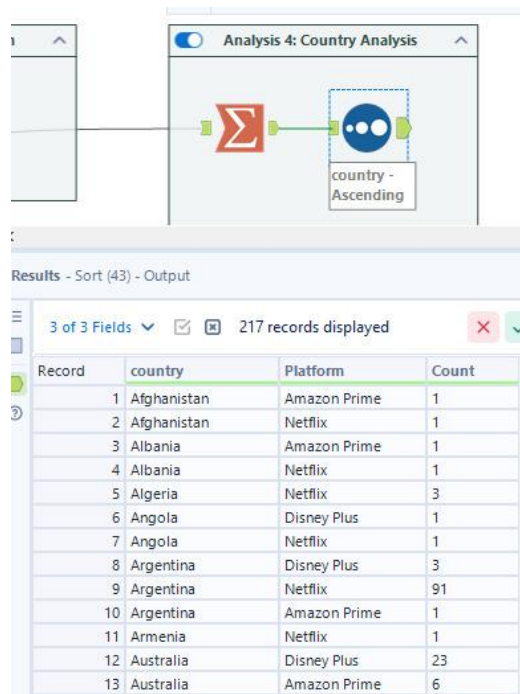
Genre analysis helps identify the content specialization of each platform and highlights the most common genres available to viewers.

Analysis 4: Country Analysis

This analysis investigates the geographical distribution of content across different countries.

Method:

- Country standardization
- Group By Country
- Group By Platform
- Count Title



Analysis 4: Country Analysis

Results - Sort (43) - Output

3 of 3 Fields 217 records displayed

Record	country	Platform	Count
1	Afghanistan	Amazon Prime	1
2	Afghanistan	Netflix	1
3	Albania	Amazon Prime	1
4	Albania	Netflix	1
5	Algeria	Netflix	3
6	Angola	Disney Plus	1
7	Angola	Netflix	1
8	Argentina	Disney Plus	3
9	Argentina	Netflix	91
10	Argentina	Amazon Prime	1
11	Armenia	Netflix	1
12	Australia	Disney Plus	23
13	Australia	Amazon Prime	6

Figure 5: Analysis 4 – Country Analysis

Interpretation:

Country analysis identifies which countries contribute the largest number of titles and provides insight into the global content strategies of streaming platforms.

Analysis 5: Trend Analysis

This analysis evaluates content trends based on release year.

Method:

- Group By Platform
- Group By Release Year
- Count Title

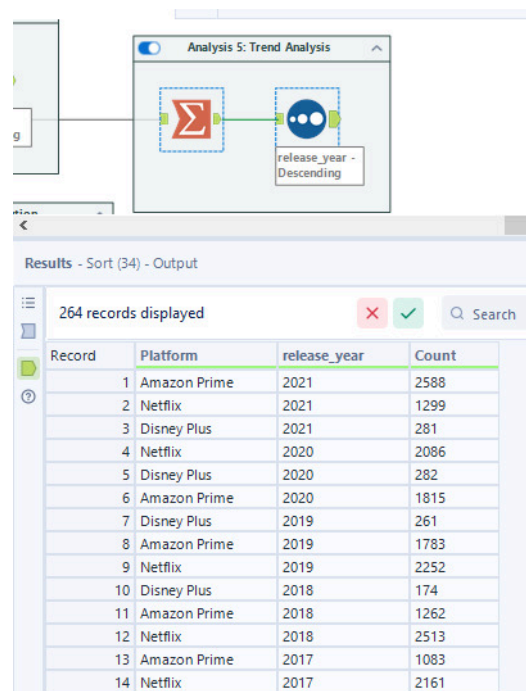


Figure 6: Analysis 5 – Trend Analysis

Interpretation:

Trend analysis provides insight into content growth and production patterns over time. It helps identify periods of rapid content expansion and changes in content strategy.

Analysis 6: Rating Analysis

This analysis examines audience rating distributions across streaming platforms.

Missing and inconsistent rating values were standardized by replacing null or empty values with “Unknown”.

Method:

- Rating standardization
- Group By Platform
- Group By Rating
- Count Title

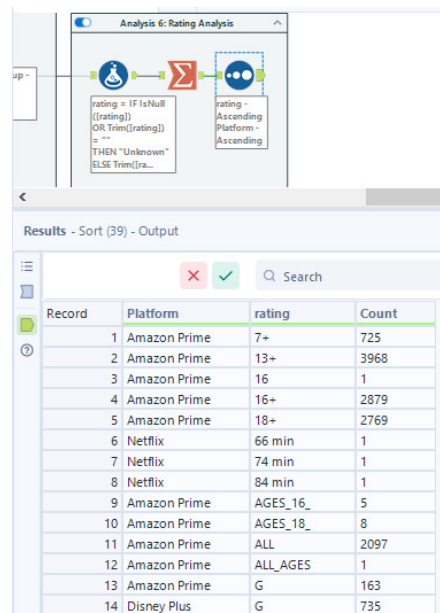


Figure 7: Analysis 6 – Rating Analysis

Interpretation:

The rating distribution helps identify the target audience segments of each platform and shows whether platforms focus more on family-friendly or mature content.

Conclusion

Netflix, Amazon Prime Video, and Disney+ datasets were successfully integrated, cleaned, processed, and analyzed by the Alteryx workflow. Reliable analysis was made possible by data preparation procedures that enhanced the datasets' consistency and quality.

Platform comparison, content type analysis, genre analysis, nation analysis, trend analysis, and rating analysis are the six analytical workflows that were created to support the project objectives. These workflows' outputs offer a solid basis for Phase 3 Power BI dashboard construction and business insight production.